Notes: Peters (Econometrica, 2020)

Heterogeneous Markups, Growth, and Endogenous Misallocation

Contents

1 Big picture

- **Question.** Can the distribution of markups be derived as an equilibrium outcome of firm dynamics rather than imposed exogenously, and what does that imply for misallocation, labor shares, and growth?
- **Core idea.** Firms build market power gradually. In a given product, a producer starts with a low markup because Bertrand competition forces limit pricing. If it keeps improving its own productivity, it pulls away from rivals and raises its markup. When a rival replaces it through creative destruction, the markup is reset.
- **State variable that matters.** The key endogenous statistic is the *churning intensity*, the rate of creative destruction relative to the rate at which incumbents accumulate market power through own-innovation. This single object shapes the whole equilibrium distribution of markups.
- **Main theoretical result.** In stationary equilibrium, markups follow an endogenous Pareto distribution. Faster churning compresses the distribution, lowers average markups, raises the labor share, and reduces allocative inefficiency.
- **Why this is different from standard misallocation papers.** The usual Hsieh-Klenow style exercise treats firm-level wedges as primitive. Peters instead makes one wedge — monopoly power — endogenous, ties it to innovation and creative destruction, and studies how policies or frictions move both growth and misallocation at the same time.
- **Why firm dynamics matter.** Growth comes from three sources: entry by new firms, expansion by incumbents into new products, and productivity improvements on a firm's existing products. The third margin is novel in this framework and is the source of rising markups inside surviving product lines.
- **Empirical application.** The model is calibrated to Indonesian manufacturing plant data. The data show that markups rise with age and that firms also grow over the life cycle, exactly the pair of facts the model is built to explain.
- **Quantitative conclusion.** Markups plausibly explain only a limited share of measured revenue-productivity dispersion in Indonesia and imply a modest static TFP loss. But frictions that keep incumbents from expanding into new product markets can substantially shrink average firm size and increase misallocation.
- **Main economic lesson.** Creative destruction is pro-competitive. A decline in churn lets surviving firms accumulate market power, which raises markup dispersion and lowers allocative efficiency.

2 Data and measurement (institutional and quantitative objects)

2.1 Theoretical objects the paper wants to connect to data

- **Product-level markup.** A product's markup is the productivity advantage of the leader relative to the follower. This is the primitive object behind misallocation in the model.
- **Firm-level markup.** A firm's markup is the harmonic mean of its product-level markups. This matters because the data observe firms or plants, not individual product lines.

- **Churning intensity.** The theory maps the whole distribution of markups into the ratio of creative destruction to own-innovation. Empirically, the life cycle of markups and firm size is informative about those two forces.
- **Misallocation wedges.** Aggregate outcomes depend on two sufficient statistics: a TFP wedge M and a labor-share wedge Λ, both functions of the endogenous markup distribution.

2.2 Indonesian manufacturing data

- **Source.** The empirical application uses the *Statistik Industri*, the annual census of large and medium-sized manufacturing firms in Indonesia.
- **Sample period.** The paper focuses on the pre-crisis period, 1990–1998.
- **Coverage.** The sample contains about 180,000 firm-year observations and includes firms with at least 20 employees.
- **Variables.** The data include revenue, employment, capital, intermediate inputs, and panel identifiers, which are enough to construct size dynamics, entry and exit, and factor-share-based markup measures.
- **Selection issue.** The census excludes most micro-firms. That is a limitation for measuring true entry and exit, but it also means the paper focuses on the segment where strategic pricing, productivity investment, and multi-product expansion are most plausible.

2.3 Descriptive moments from Indonesia

- **Firm size distribution.** The average plant has about 143 employees, the median has 45, the 25th percentile has 27, and the 90th percentile has 351.
- **Entry and exit.** The average entry rate in the census is 10.4%, and the exit rate is 8.2%.
- **Contribution of entrants and exiters.** Entering firms account for about 5.0% of employment and 3.9% of sales, while exiting firms account for 4.4% of employment and 3.3% of sales.
- **Interpretation.** New and disappearing firms are small relative to incumbents, which is consistent with the model's view that entrants start with low markups and small scale.

2.4 Markup measurement in the data

- **Underlying accounting identity.** The paper uses the De Loecker-Warzynski logic that, for a variable input,

  $$\mu_f = \alpha_{\ell,f}\, s_{\ell,f}^{-1}, \tag{1}$$

  where $\alpha_{\ell,f}$ is the output elasticity of labor and $s_{\ell,f}$ is labor's share in value added.
- **Why the level is not essential here.** The paper mostly needs markup *growth over the life cycle*, not the absolute level. If labor elasticities are roughly stable within narrow industries, relative changes in inverse labor shares are informative about relative changes in markups.

- **Baseline empirical proxy.** The paper estimates

$$\ln s_{\ell,ft}^{-1} = \delta_s + \delta_t + u_{ft}, \tag{2}$$

and takes the residual $\hat{u}_{ft}$ as the firm's relative log markup.

- **Robustness.** The paper also controls for capital intensity, uses a balanced panel, and constructs markup proxies from material shares. The life-cycle pattern is robust across these alternatives.

2.5 Moments used for calibration

- **Markup life cycle.** The average log markup of a 7-year-old firm is about 0.082 log points above that of an entrant.
- **Employment life cycle.** The average log employment of a 7-year-old firm is about 0.494 log points above that of an entrant.
- **Entry rate.** The targeted entry rate is 10.4%.
- **Aggregate growth.** The targeted aggregate productivity growth rate is 3%.
- **Role of these moments.** These four moments map directly into the endogenous innovation rates (I, x, z) and the step size λ, after which the structural innovation-cost parameters are backed out from equilibrium conditions.

3 Models and empirical specifications (core equations + interpretation)

3.1 Preferences, product markets, and production

Households have log utility:

$$U = \int_0^\infty e^{-\rho t} \ln(c_t)\, \mathrm{d}t. \tag{3}$$

The final good is a Cobb-Douglas aggregate across a continuum of product markets, with perfect substitutability across firms within a product market. The exact within-market allocation is not the key object; what matters is that sales are equalized across products in equilibrium.

A firm with product-specific productivity q_{fi} produces

$$y_{fi} = q_{fi}\ell. \tag{4}$$

- **Economic meaning.** Different firms can produce the same product. The only firm heterogeneity is product-level productivity.
- **Why this matters.** Constant returns and Bertrand competition make the follower's marginal cost the key determinant of the leader's markup.

3.2 Static allocation: markups, employment, and profits

Because firms compete à la Bertrand within a product market, only the most productive producer is active. Let q_i denote the leader's productivity and q_i^F the follower's productivity. The product-level markup is

$$\mu_i \equiv \frac{p_i}{w/q_i} = \frac{w/q_i^F}{w/q_i} = \frac{q_i}{q_i^F}. \tag{5}$$

- **Interpretation.** Markups are entirely pinned down by the leader's productivity advantage over the next best rival.
- **Key implication.** Higher productivity does not automatically mean a higher markup; what matters is relative productivity inside the market.

The equalization of sales across products implies product-level labor demand and profits:

$$\ell_i = \frac{1}{q_i} y_i = \frac{1}{q_i}\frac{Y}{p_i} = \mu_i^{-1}\frac{Y}{w}, \qquad \pi_i = \left(1 - \mu_i^{-1}\right) Y. \tag{1}$$

- **Interpretation.** Higher markups reduce labor hired in a product line and increase profits.
- **Misallocation channel.** A high-markup firm undersupplies employment relative to the efficient benchmark.

At the firm level, if N_f is the set of products and $n_f = |N_f|$, employment is

$$\ell_f = \sum_{i \in N_f} \ell_i = \frac{Y}{w} n_f \mu_f^{-1}, \qquad \mu_f = \left(\frac{1}{n_f}\sum_{i \in N_f} \mu_i^{-1}\right)^{-1}. \tag{2}$$

- **Interpretation.** Firm size depends on two forces: how many products the firm sells and how high its average markup is.
- **Important subtlety.** Multi-product firms can have firm-level markups that are much smoother than the product-level markups that actually drive allocative efficiency.

3.3 Aggregation and the static cost of monopoly power

Let $G_t(\mu)$ denote the cross-sectional distribution of product-level markups. Aggregate output can be written as

$$Y_t = Q_t M_t L_{Pt}, \qquad M_t \equiv \frac{\exp(\int \ln \mu^{-1} \, \mathrm{d}G_t(\mu))}{\int \mu^{-1} \, \mathrm{d}G_t(\mu)}. \tag{3}$$

The labor share is

$$\frac{w_t L_{Pt}}{Y_t} = \Lambda_t, \qquad \Lambda_t \equiv \int \mu^{-1} \, \mathrm{d}G_t(\mu). \tag{4}$$

- M_t**.** This is the allocative-efficiency wedge. If all markups are equal, then $M_t = 1$. Dispersion in markups pushes M_t below one.
- Λ_t**.** This is the labor-share wedge. Average markups lower labor's share even if there is no markup dispersion.
- **Key insight.** The model separates the effect of monopoly power on TFP (through dispersion) from its effect on factor shares (through the average markup).

3.4 Dynamics: quality ladders, own-innovation, and creative destruction

Productivities live on a quality ladder:

$$q_{r+1} = \lambda q_r. \tag{6}$$

If the leader is Δ_{it} rungs above the follower, then

$$\mu_{it} = \frac{q_{it}}{q_{it}^F} = \lambda^{\Delta_{it}}. \tag{5}$$

- **Three innovation margins.**
  - New entrants can replace incumbents in a market.
  - Existing firms can expand into a new market and replace the incumbent there.
  - The current producer can improve its own product and widen its productivity lead.
- **Why the third margin is crucial.** Klette-Kortum-style models already have entry and expansion. Peters adds own-innovation on existing products, which creates the gradual accumulation of market power.

3.5 Firm problem and innovation technology

Let $V_t(n, [\Delta_i])$ denote the value of a firm with n products and product-level quality gaps $[\Delta_i]$. The Bellman equation can be written compactly as

$$\begin{aligned} r_t V_t(n, [\Delta_i]) = \sum_{j=1}^{n} \pi_t(\Delta_j) + \dot{V}_t(n, [\Delta_i]) - \sum_{j=1}^{n} \tau_t \Big[V_t(n, [\Delta_i]) - V_t(n-1, [\Delta_i]_{i \neq j}) \Big] \\ + \max_{[x_j, I_j]} \Bigg\{ \sum_{j=1}^{n} I_j \Big[V_t(n, [\Delta_i]_{i \neq j}, \Delta_j + 1) - V_t(n, [\Delta_i]) \Big] \\ + \sum_{j=1}^{n} x_j \Big[V_t(n+1, [\Delta_i], 1) - V_t(n, [\Delta_i]) \Big] - \mathcal{I}([x_i, I_i]; n, [\Delta_i]) w_t \Bigg\}. \end{aligned} \tag{7}$$

- **Terms in order.** Current profits; capital gain; risk of losing existing products through creative destruction; option value of own-innovation and expansion; cost of research activity.
- **Creative destruction rate.** Each product can be replaced at the endogenous rate τ_t.

The innovation cost function is additively separable across products:

$$\mathcal{I}([x_i, I_i]; n, [\Delta_i]) = \sum_{i=1}^{n} c(I_i, x_i; \Delta_i), \qquad c(I, x; \Delta) = \lambda^{-\Delta} \frac{1}{\phi_I} I^{\zeta} + \frac{1}{\phi_x} x^{\zeta}. \tag{8}$$

Potential entrants have a linear entry technology. Under positive entry, the free-entry condition is

$$V_t(1, 1) = \frac{w_t}{\phi_z}. \tag{9}$$

The rate of creative destruction is

$$\tau_t = z_t + \int x_{it} \, \mathrm{d}i. \tag{10}$$

- **ϕ_I, ϕ_x, ϕ_z.** These are the efficiency shifters for own-innovation, incumbent expansion, and entry.
- **$\zeta > 1$.** Convexity ensures a unique optimal innovation choice.
- **Interpretation of barriers.** Lower ϕ_x or ϕ_z can be read as higher technological or institutional barriers to expansion and entry.

3.6 Stationary equilibrium and closed-form policies

In balanced growth, the value function separates neatly:

$$V_t(n, [\Delta_i]) = \sum_{i=1}^{n} V_t(\Delta_i) = V_t^P n + \sum_{i=1}^{n} V_t^M(\Delta_i), \tag{6}$$

with

$$V_t^P = \frac{\pi_t(1) + (\zeta - 1)\dfrac{x^\zeta}{\phi_x} w_t}{\rho + \tau}, \qquad V_t^M(\Delta) = \frac{\pi_t(\Delta) - \pi_t(1) + (\zeta - 1)\lambda^{-\Delta}\dfrac{I^\zeta}{\phi_I} w_t}{\rho + \tau}. \tag{11}$$

The optimal expansion rate is

$$x = \left(\frac{\phi_x}{\zeta \phi_z}\right)^{1/(\zeta-1)}. \tag{7}$$

The optimal own-innovation rate solves

$$I = \left[\frac{\lambda - 1}{\lambda} \frac{1}{\rho + \tau} \frac{\phi_I}{\zeta} \frac{Y_t}{w_t} - \frac{\zeta - 1}{\zeta} I^\zeta\right]^{1/(\zeta-1)}. \tag{8}$$

Aggregate growth is

$$g = \frac{\dot{Q}_t}{Q_t} = \ln \lambda \cdot (I + x + z) = \ln \lambda \cdot (I + \tau). \tag{12}$$

- **Interpretation of V_t^P.** This is the value of simply controlling a product and using it as a platform for expansion.
- **Interpretation of $V_t^M(\Delta)$.** This is the extra value of market power created by a productivity lead.
- **What lowers own-innovation.** A higher τ shortens the expected duration of monopoly rents, so it lowers I.
- **Growth accounting in the model.** Growth comes from the sum of own-innovation and all forms of creative destruction.

3.7 Distribution of markups and the main theorem

Let $\nu_t(\Delta)$ be the mass of products with quality gap Δ. Its law of motion is

$$\dot{\nu}_t(\Delta) = \begin{cases} -(\tau + I)\nu_t(\Delta) + I\nu_t(\Delta - 1), & \Delta \geq 2, \\ \tau(1 - \nu_t(1)) - I\nu_t(1), & \Delta = 1. \end{cases} \tag{9}$$

In stationary equilibrium,

$$\nu(\Delta) = \frac{\vartheta}{(1+\vartheta)^{\Delta}}, \qquad \vartheta \equiv \frac{\tau}{I}. \tag{13}$$

Define

$$\theta = \frac{\ln(1+\vartheta)}{\ln\lambda}. \tag{14}$$

Then the stationary markup distribution is Pareto:

$$G(\mu) = 1 - \mu^{-\theta}. \tag{15}$$

The aggregate wedges are

$$M = e^{-1/\theta}\frac{1+\theta}{\theta}, \qquad \Lambda = \frac{\theta}{1+\theta}. \tag{10}$$

- **Main theorem.** The whole equilibrium distribution of markups is summarized by one endogenous statistic, $\vartheta = \tau/I$.
- **Economic meaning.** More creative destruction relative to own-innovation means firms have less time to build market power, so the right tail of markups shrinks.
- **Comparative static.** M and Λ are increasing in ϑ. Faster churn raises allocative efficiency and the labor share.

3.8 Life-cycle implications for products and firms

At the product level, expected log markups rise with product age:

$$\mathbb{E}[\ln\mu \mid \text{product age} = a^P] = \ln\lambda\,(1 + Ia^P). \tag{11}$$

- **Interpretation.** Conditional on survival, a product line keeps ratcheting up its markup as the incumbent innovates on top of its own past success.

At the firm level, the average markup of a firm of age a_f is

$$\mathbb{E}[\ln\mu_f \mid \text{firm age} = a_f] = \ln\lambda\Big(1 + I\mathbb{E}[a^P \mid a_f]\Big). \tag{12}$$

Firm survival is

$$S(a) = 1 - \frac{\tau}{x}\gamma(a), \tag{13}$$

where $\gamma(a)$ is a closed-form function of x and τ given in the paper.

Finally, the life cycle of firm employment is

$$\mathbb{E}[\ln\ell_f \mid a_f] - \mathbb{E}[\ln\ell_f \mid 0] = \underbrace{\big(1-\gamma(a)\big)\sum_{j=1}^{\infty}\ln j\cdot\gamma(a)^{j-1}}_{\text{sales growth from adding products}} - \underbrace{\ln\lambda\cdot\big(I\mathbb{E}[a^P \mid a_f]\big)}_{\text{employment loss from rising markups}}. \tag{14}$$

- **Two opposing forces.** Firms grow by adding products, but they also become more distorted by raising markups on older products.
- **Empirical implication.** A positive slope in both the employment life cycle and the markup life cycle implies that both horizontal expansion and own-innovation matter.

3.9 Extensions

Stochastic step size. If an innovation jumps by j rungs with probability p_j, then the stationary quality-gap distribution becomes

$$\nu(\Delta) = \frac{1}{1+\vartheta} \sum_{m=1}^{\Delta-1} \nu(m) p_{\Delta-m} + \frac{\vartheta}{1+\vartheta} p_{\Delta}. \tag{16}$$

If $p_j = (1-\kappa)\kappa^{j-1}$, then markups are again Pareto with

$$\theta(\kappa) = \frac{1}{\ln \lambda} \ln\left(\frac{1+\vartheta}{1+\kappa\vartheta}\right). \tag{17}$$

- **Takeaway.** The qualitative role of churning is unchanged. A higher ϑ still compresses the distribution and lowers misallocation.

CES demand. With elasticity of substitution $\sigma > 1$, the markup becomes

$$\mu(\Delta) = \min\left\{\frac{\sigma}{\sigma-1}, \lambda^{\Delta}\right\}. \tag{18}$$

The aggregate wedges are then

$$M = \frac{\left(\int \mu(\Delta)^{1-\sigma} \left(\frac{q}{Q}\right)^{\sigma-1} \, \mathrm{d}F(q,\Delta)\right)^{\sigma/(\sigma-1)}}{\int \mu(\Delta)^{-\sigma} \left(\frac{q}{Q}\right)^{\sigma-1} \, \mathrm{d}F(q,\Delta)}, \tag{19}$$

$$\Lambda = \frac{\int \mu(\Delta)^{-\sigma} \left(\frac{q}{Q}\right)^{\sigma-1} \, \mathrm{d}F(q,\Delta)}{\int \mu(\Delta)^{1-\sigma} \left(\frac{q}{Q}\right)^{\sigma-1} \, \mathrm{d}F(q,\Delta)}. \tag{20}$$

- **Takeaway.** CES demand complicates the algebra because the joint distribution of productivity and markups matters. But the main mechanism survives: creative destruction remains pro-competitive.

4 Identification, calibration, and credibility

4.1 What identifies the model?

- **This is not an IV paper.** There is no quasi-experimental design. Identification comes from the model's closed-form mapping from life-cycle moments and entry rates to the innovation forces (I, x, z) and the step size λ.
- **Markup life cycle.** A steeper age-markup profile requires stronger own-innovation relative to churning and helps pin down I and λ.
- **Employment life cycle.** Faster life-cycle growth in employment points to more incumbent expansion x relative to markup accumulation.

- **Entry rate.** This disciplines the balance between entrant creative destruction and incumbent-driven creative destruction.
- **Aggregate productivity growth.** This pins down the product of innovation intensity and step size through $g = \ln \lambda(I + \tau)$.

4.2 Benchmark calibration

- **Externally fixed parameters.** The paper sets $\zeta = 2$ and $\rho = 0.05$.
- **Calibrated structural parameters.** The benchmark implies $\phi_I = 8.896$, $\phi_x = 1.077$, $\phi_z = 2.590$, and $\lambda = 1.038$.
- **Implied endogenous rates.** The equilibrium innovation rates are $I = 0.561$, $x = 0.207$, $z = 0.032$, and therefore $\tau = 0.240$.
- **Immediate implication.** Own-innovation is quantitatively important in the calibrated model, but creative destruction still happens fast enough to keep the markup distribution fairly tight.

4.3 What makes the quantification credible?

- **Strong analytical structure.** Because the model yields closed-form expressions for the target moments, the calibration is transparent rather than black-box.
- **Non-targeted validation.** The model is checked against the full age profiles of markups and employment, firm survival, and the firm-size distribution.
- **Robustness.** The paper shows that the misallocation results are not very sensitive to stochastic step sizes, alternative demand elasticities, or alternative markup measurement choices.

4.4 Main limitations

- **Selected firm sample.** The Indonesian census covers large and medium firms only, so measured entry and exit are conditional on crossing the sample threshold.
- **Markup measurement.** Markups are inferred indirectly from factor shares, not observed directly.
- **Functional form discipline.** The baseline model relies on log utility over the final good, Bertrand competition, a quality ladder, and convex innovation costs. These choices buy tractability but also narrow the model's scope.
- **Static importance of markups.** Because the calibrated markup dispersion is modest, the theory implies modest TFP losses from markups alone. That is a substantive result, but it also means the model cannot explain the full size of cross-country misallocation estimates by itself.

5 Tables and figures

5.1 Figure 1 (p. 14): The life-cycle of markups

Read:

- The theoretical firm-level markup profile rises early in life and then eventually flattens and slightly bends down for very old firms.
- The intuition is selection plus portfolio composition. Young firms start with low-markup products. As they survive, they accumulate market power on existing products, but they also add new low-markup products and lose old ones through creative destruction.
- The red dashed asymptote, $\ln \lambda(1 + I/\tau)$, shows the average markup of very old firms. It is lower when churning τ is high relative to own-innovation I.
- This figure is the theoretical backbone for the empirical exercise that estimates markup growth by firm age.

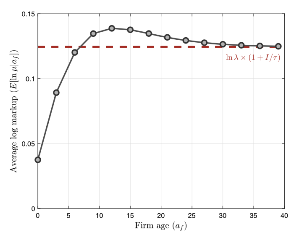

FIGURE 1.—THE LIFE-CYCLE OF MARKUPS. *Notes*: The figure displays the expected log markup by age, that is, $E[\ln \mu_f | a_f]$ given in (12).

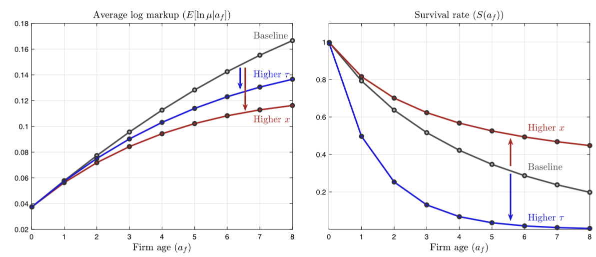

FIGURE 2.—THE LIFE-CYCLE OF MARKUPS AND FIRM SURVIVAL. *Notes*: The figure displays the expected log markup as a function of age (see (12)) in the left panel and the cumulative survival probability (see (13)) in the right panel. The baseline model is shown in gray. The red (blue) line refers to an increase in incumbent expansion x (creative destruction τ).

5.2 Figure 2 (p. 15): The life-cycle of markups and firm survival

Read:

- The left panel compares two comparative statics: higher incumbent expansion x and higher creative destruction τ both flatten the markup life cycle.
- But they do so for different reasons. A higher τ shortens the time firms can keep a product before being replaced. A higher x dilutes average firm markups because firms add more new products with low markups.
- The right panel shows the crucial difference: higher x raises firm survival by making firms broader, while higher τ sharply lowers survival.
- This is why the same observed markup life cycle can mean very different underlying dynamics depending on what happens to survival and firm growth.

5.3 Table I (p. 20): The manufacturing sector in Indonesia

Read:

- The typical formal manufacturing firm is not large by U.S. standards. The median plant has only 45 workers.
- Entry and exit rates are substantial, but entrants and exiters are small, especially in terms of sales. This is exactly what the model predicts when new firms begin with low markups and small product portfolios.

- The table makes clear that the Indonesian formal sector is thin and skewed toward small producers, which motivates the counterfactual comparison to the U.S.

TABLE I

THE MANUFACTURING SECTOR IN INDONESIA[a]

Firm Size Distribution				Entrants			Exiting Firms		
	Quantiles				Share of			Share of	
Mean	25%	50%	90%	Entry Rate	Empl.	Sales	Exit Rate	Empl.	Sales
143	27	45	351	10.4%	5.0%	3.9%	8.2%	4.4%	3.3%

[a] The table contains descriptive statistics on the sample of manufacturing plants in Indonesia. Columns 1–4 contain selected statistics about the distribution of employment. Columns 5 and 8 contain the entry and exit rate. Columns 6 and 9 (7 and 10) report the employment (sales) share of entering and exiting firms. All results are simple averages over the time period of the sample.

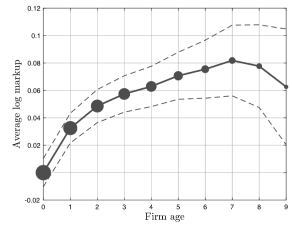

FIGURE 3.—THE LIFE-CYCLE OF MARKUPS IN INDONESIA. *Notes*: I focus on the unbalanced panel of firms entering the economy after 1990. I calculate log markups within 5-digit-industry-year cells, then calculate the average by the age of the cohort and normalize log markups of entering cohorts to zero. Because of attrition, the size of the cohort is declining in age. The dots reflect the size of the cohort. I also depict the 90% confidence intervals around the estimated average profile.

5.4 Figure 3 (p. 22): The life-cycle of markups in Indonesia

Read:

- The empirical age-markup profile rises steadily for roughly the first seven years, by about 0.08 log points.
- This is the central reduced-form fact supporting the model's own-innovation channel. In a pure Klette-Kortum environment with no own-innovation on existing products, markups would not rise with age in this way.
- The confidence bands widen for older ages because cohort attrition reduces the number of surviving firms.

5.5 Table II (p. 23): The life-cycle of markups in Indonesia

Read:

- The coefficient on age is about 1.0% to 1.4% per year, depending on controls.
- The result survives controls for capital intensity, nonparametric controls for the capital-labor ratio, and conditioning on the balanced panel.
- So the age-markup profile is not just a by-product of changing factor intensities or of selective firm exit.

5.6 Table III (p. 25): Calibration

Read:

- The benchmark calibration exactly matches the four targeted moments: the markup life cycle, employment life cycle, entry rate, and aggregate growth rate.
- The implied own-innovation rate $I = 0.561$ is large relative to the entrant flow $z = 0.032$, while incumbent expansion is also quantitatively important at $x = 0.207$.
- The implied rate of creative destruction is $\tau = x + z = 0.240$.

TABLE II

THE LIFE-CYCLE OF MARKUPS IN INDONESIA[a]

	Dependent Variable: Log Markup			
Age	0.0137	0.0103	0.0106	0.0108
	(0.00122)	(0.00117)	(0.00116)	(0.00127)
Industry FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Control for k/l		$\ln k/l$	non-param.	$\ln k/l$
Balanced panel				✓
N	55,212	55,212	55,212	42,434
R^2	0.219	0.287	0.305	0.298

[a]Robust standard errors in parentheses. I focus on the unbalanced panel of firms that enter the market after 1990. I use the data from 1991 to 2000. Columns 2 and 4 control for the log capital-labor ratio $\ln k/l$. Column 3 controls non-parametrically by including 50 fixed effects for 50 quantiles of $\ln k/l$. Column 4 focuses on the balanced panel. Industry fixed effects control for industry affiliation at the 5-digit level.

TABLE III

CALIBRATION: MOMENTS AND STRUCTURAL PARAMETERS[a]

Targets	Data	Model	Calibrated Parameters			Outcomes	
Markup life-cycle	0.082	0.082	φ_I	Innovation efficiency	8.896	I	0.561
Empl. life-cycle	0.494	0.494	φ_x	Expansion efficiency	1.077	x	0.207
Entry rate	10.4%	10.4%	φ_z	Entry efficiency	2.590	z	0.032
Agg. growth	3%	3%	λ	Step size	1.038	τ	0.240
–	–	–	ζ	Cost function	2		
–	–	–	ρ	Discount rate	0.05		

[a]This table contains the calibrated parameters, the targeted data moments, the resulting moments in the model, and the equilibrium outcomes for I, x, and z. Given (I, x, z), all other equilibrium allocations can be calculated. I measure the life-cycle of markups and employment as the log difference between entrants and 7-year-old firms. The two parameters ζ and ρ are set exogenously. As explained in the text, the mapping from the data moments to (I, x, z, λ) does not depend on ζ and ρ.

- The step size is modest, $\lambda = 1.038$, which helps explain why average markups in the calibrated model are not huge.

5.7 Table IV (p. 25): Sensitivity matrix

Read:

- A 5% increase in innovation efficiency ϕ_I strongly raises own-innovation and steepens the markup life cycle while flattening employment growth.
- A 5% increase in expansion efficiency ϕ_x raises incumbent creative destruction and steepens employment growth while flattening the markup life cycle.
- A 5% increase in entry efficiency ϕ_z mainly works through entry and overall churn.
- This table makes the identification logic concrete: different moments move in different directions with ϕ_I, ϕ_x, ϕ_z, and λ.

TABLE IV

SENSITIVITY MATRIX[a]

	Change in …				Initial Level
	φ_I	φ_x	φ_z	λ	
Effects on Endogenous Outcomes					
Rate of own-innovation (I)	4.0%	−1.1%	−0.9%	1.9%	0.561
Incumbent creative destruction (x)	0.0%	5.0%	−4.8%	0.0%	0.2078
Entry (z)	8.1%	−21.9%	38.1%	24.3%	0.0326
Creative destruction (τ)	1.1%	1.4%	1.1%	3.3%	0.2404
Effects on Equilibrium Moments					
Life-cycle of markups	3.6%	−3.3%	0.6%	5.8%	0.0818
Life-cycle of employment	−1.5%	6.9%	−7.9%	−3.5%	0.494
Entry rate	3.5%	−7.1%	12.9%	10.2%	0.104
Growth rate	3.1%	−0.3%	−0.3%	7.3%	0.03

[a]The table reports the effect of a 5% change in the relative innovation efficiencies $(\varphi_I, \varphi_x, \varphi_z)$ and the quality increase $(\lambda - 1)$ on the endogenous outcomes (top panel) and the equilibrium moments (lower panel).

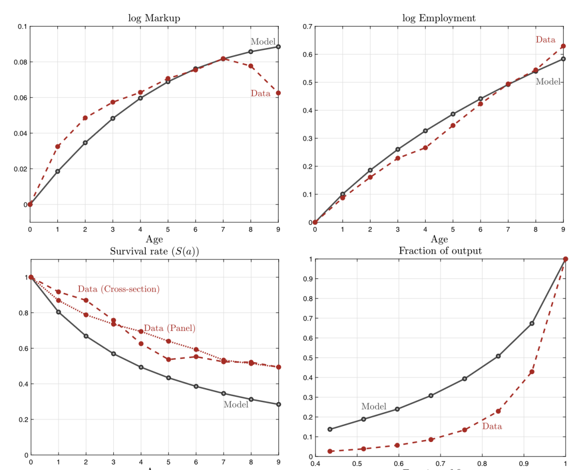

FIGURE 4.—NON-TARGETED MOMENTS: MODEL VERSUS DATA. *Notes*: The figure compares the model and the data for the life-cycle of markups (top-left panel), employment growth (top-right panel), survival (bottom-left panel), and the concentration of value added (bottom-right panel). Firm survival is measured both by the share of firms by age in the 2000 cross-section and by the share of firms of the entering cohort in 1991 by age. For the employment life-cycle, see Section A-2.3 in the Supplemental Material. To measure the concentration of value added, I drop the highest and lowest 3% of firms.

5.8 Figure 4 (p. 26): Non-targeted moments — model versus data

Read:

- The model tracks the broad shape of the observed markup and employment life cycles even though it targets only the age-7 values.
- It does a reasonable job on firm survival, though it predicts somewhat more shakeout than seen in the data.
- It underpredicts concentration in the firm-size distribution. That suggests the model is missing some persistent heterogeneity that makes large firms both larger and more dominant than the baseline theory allows.

5.9 Table V (p. 28): Markups and misallocation in Indonesia

Read:

- The endogenous Pareto tail parameter is $\theta = 9.5$.
- Average markups are about 11.8%, and the standard deviation of log markups is 0.103 across products but only 0.079 across firms.
- The misallocation wedge is $M = 0.995$, so markups alone reduce TFP by about 0.5% in the benchmark.
- The labor wedge is $\Lambda = 0.887$, so monopoly power depresses labor's share more than it depresses TFP.
- The broader lesson is that even moderate markup dispersion can have a visible distributional effect, but the static TFP effect is quantitatively limited here.

TABLE V

MARKUPS AND MISALLOCATION IN INDONESIA[a]

Markups and Misallocation					
θ	$E[\mu]$	$\sigma(\ln \mu)$	$\sigma(\ln \mu_f)$	$\mathcal{M}$	Λ
9.5	11.8%	0.103	0.079	0.995	0.9

[a]The table reports the endogenous tail parameter of the markup distribution θ, the average markup ($E[\mu]$), the dispersion of log markups across products ($\sigma(\ln \mu)$) and firms ($\sigma(\ln \mu_f)$), and the two misallocation wedges $\mathcal{M}$ and Λ (see (3) and (4)).

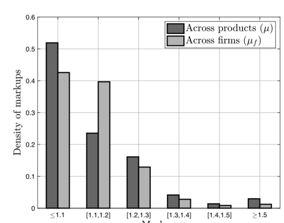

FIGURE 5.—THE STATIONARY DISTRIBUTION OF MARKUPS. *Notes*: The figure shows the distribution of markups at the product level (dark bars) and at the firm level (light bars). The results are based on the calibration reported in Table III.

5.10 Figure 5 (p. 29): The stationary distribution of markups

Read:

- The dark bars show the product-level markup distribution; the light bars show the firm-level distribution.
- The firm-level distribution is noticeably more compressed because a multi-product firm's observed markup averages across high- and low-markup product lines.

- This is an important warning for empirical misallocation work: firm-level dispersion in TFPR or markups can understate the underlying product-level dispersion that matters for welfare.

5.11 Figure 6 (p. 30): Markups and misallocation under stochastic step size and CES demand

Read:

- The left panel shows that once the model is recalibrated, allowing stochastic innovation step sizes changes misallocation very little.
- The right panel shows that moving from Cobb-Douglas to CES demand raises the implied TFP losses from markup dispersion, but not by a dramatic amount. Even at higher elasticities, the losses remain modest.
- So the headline quantitative conclusion — markups matter, but not enough to explain large cross-country TFP gaps by themselves — is robust.

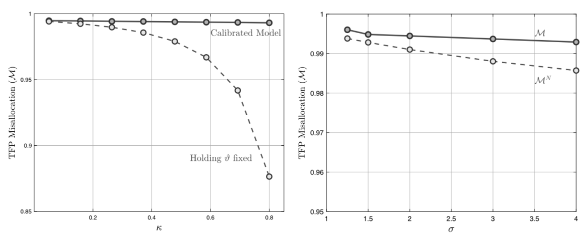

FIGURE 6.—MARKUPS AND MISALLOCATION: STOCHASTIC STEP SIZE AND CES PREFERENCES. *Notes*: The left panel displays the misallocation wedge $\mathcal{M}$ as a function of κ. The implications for the recalibrated model are shown as the solid line. For comparison, I also depict the case where I vary only κ but keep ϑ at its baseline value. The right panel shows the misallocation wedges $\mathcal{M}$ and $\mathcal{M}^N$ as a function of the demand elasticity σ. In the dark line, I show the expressions when the model is calibrated to the same moments as the baseline model. In the dashed line, I depict the "naive" measure, which keeps the distribution of markups at their baseline level.

TABLE VI

COUNTERFACTUAL: CHANGES IN EXPANSION AND ENTRY COSTS[a]

	Heterogeneous Markups			Constant Markups
	Indonesia	U.S.	Change	Change
Calibration				
Entry rate	10.4%	8%		
Employment life-cycle	1.7	2		
Expansion costs	1	0.67	−33%	−21%
Entry costs	1	0.86	−14%	−2%
Panel A: The Distribution of Firm Size				
Number of firms (F)	0.313	0.122	−60.1%	−61%
Output share of small firms (Ω_1)	0.136	0.035	−73.9%	−73%
Panel B: Markups, Misallocation, and Growth				
Life-cycle of markups	8.2%	6.2%	0.02	–
Average markup	11.73%	9.25%	−0.025	–
Markup dispersion (products)	10.5%	8.47%	−19.3%	–
Markup dispersion (firms)	8.1%	5.86%	−27.5%	–
Aggregate misallocation ($\mathcal{M}$)	0.995	0.997	0.2%	–
Labor wedge (Λ)	0.887	0.903	1.8%	–
Growth rate (g)	3%	2.91%	−0.0009	−0.0004

[a]The first panel contains the calibration moments. The parameters for the Indonesian economy are contained in Table III. For the U.S. economy, I recalibrate the relative efficiency of expansion and entry, that is, φ_Z and φ_X. All remaining parameters are the same as in Table III. In the last column, I report the results of a calibration of a model with constant markups.

5.12 Table VI (p. 32): Counterfactual changes in expansion and entry costs

Read:

- To match U.S.-style moments, the model needs lower barriers to both expansion and entry, but the reduction in expansion costs is much bigger: roughly 33% versus 14%.
- The result is a much thinner mass of firms and much larger average firm size: the number of firms falls by about 60%, and the output share of the smallest firms falls by about 74%.
- Markups and markup dispersion decline, so allocative efficiency rises. The TFP wedge improves from 0.995 to 0.997, which means about one-third of the original markup-driven misallocation disappears.
- Growth hardly changes. More churn lowers monopoly rents and discourages own-innovation, so the growth effects largely offset.
- The constant-markup comparison in the last column shows that the importance of incumbent expansion barriers for firm size and growth is not driven entirely by markup heterogeneity.

6 Short summary

- **Conclusion.** The paper builds a tractable endogenous-growth model in which firms accumulate markups on surviving products and lose them when creative destruction arrives.
- **Intuition.** Own-innovation raises market power; churn resets it. The ratio of those two forces determines the equilibrium markup distribution.
- **Empirical lesson.** The life cycle of markups and employment contains enough information to separately discipline own-innovation, incumbent expansion, and entry.
- **Quantitative lesson.** In Indonesia, markups contribute to measured misallocation, but their static TFP cost is modest. The bigger effect of weak incumbent expansion is on firm size and the prevalence of small firms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Endogenous markups.** The paper shows that heterogeneous markups can arise endogenously from firm dynamics rather than from exogenous productivity draws alone.
- **One sufficient statistic.** The markup distribution is pinned down by churning intensity, $\vartheta = \tau/I$.
- **Macro implications.** Faster churn lowers average markups, reduces markup dispersion, raises the labor share, and improves allocative efficiency.
- **Empirical support.** Indonesian firms display rising markups with age and rising employment with age, consistent with the coexistence of own-innovation and expansion.
- **Quantitative conclusion.** Markups are not the dominant source of cross-country misallocation in the calibration, but barriers that slow incumbent expansion are important for explaining why firms in poorer economies remain small.

7.2 Economic intuition

- A firm that survives in a product market has time to improve that product, pull away from rivals, and charge a higher markup.
- Creative destruction interrupts that accumulation process. That is why competition in the model is fundamentally dynamic rather than static.
- Policies or frictions that affect who can enter new markets alter not just the size distribution of firms but also the long-run distribution of markups and the level of misallocation.

7.3 Takeaway

This paper is valuable because it unifies three literatures that often talk past one another: endogenous growth, markups, and misallocation. The model is analytically simple enough to deliver sharp comparative statics, yet rich enough to match the life cycle of markups and firm size in plant data. The main message is subtle but important: creative destruction is not only a source of growth, it is also a force that disciplines monopoly power and keeps allocative inefficiency in check.